CS471 – Web Technologies (Laboratory)	Qassim University College of Computer	Lab 2
		Django and Basic Git operations

Lab Activities:

Part 1: Django web development framework

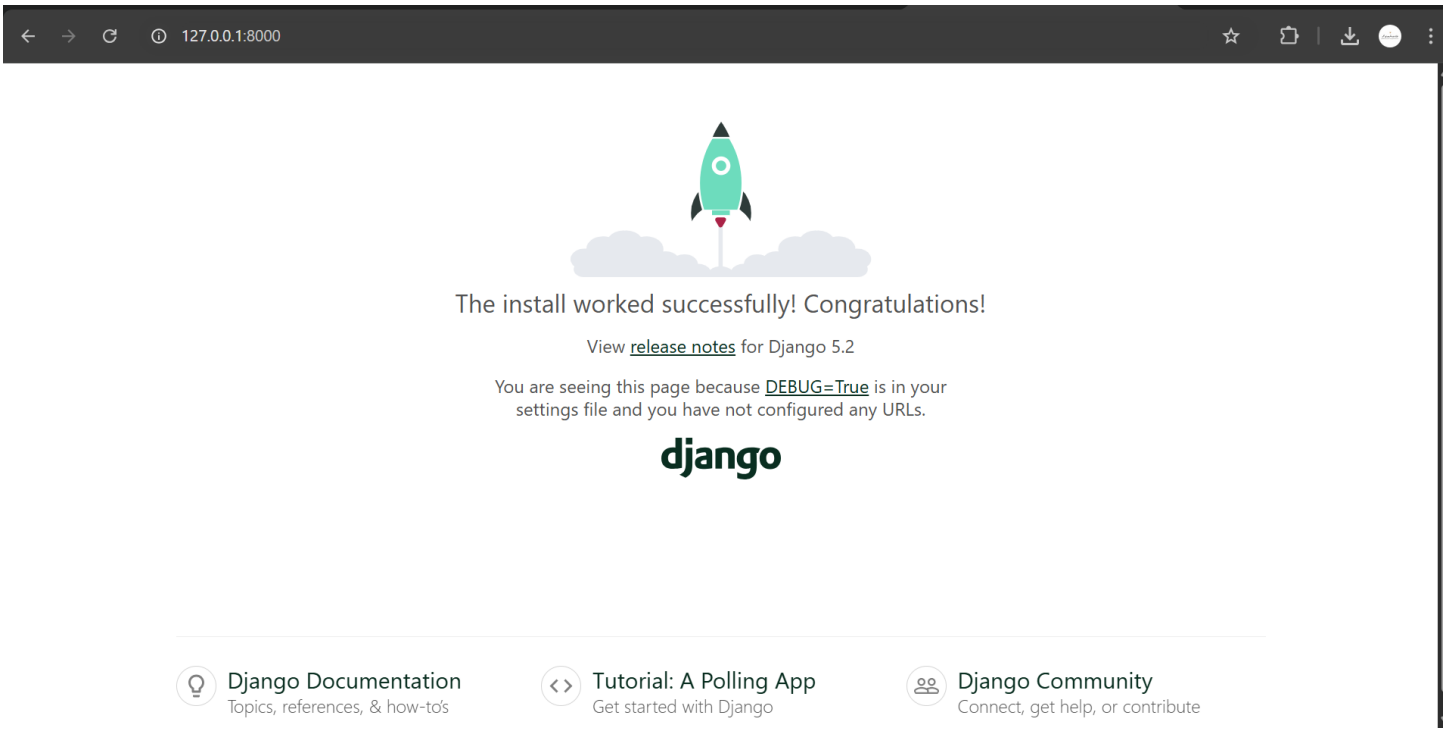
Task 1: install Django

```
PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice> python -m venv djangoenv
PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice> pip install django
WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ProtocolError('Co
nnection aborted.', RemoteDisconnected('Remote end closed connection without response'))': /simple/django/
Collecting django
  Downloading django-5.2.11-py3-none-any.whl.metadata (4.1 kB)
Collecting asgiref>=3.8.1 (from django)
  Using cached asgiref-3.11.1-py3-none-any.whl.metadata (9.3 kB)
Collecting sqlparse>=0.3.1 (from django)
  Using cached sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
Requirement already satisfied: tzdata in C:\Users\re2na\AppData\Local\Programs\Python\Python311\Lib\site-packages (from django) (2025
.3)
Downloading django-5.2.11-py3-none-any.whl (8.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 8.3/8.3 MB 365.5 kB/s  0:00:22
Using cached asgiref-3.11.1-py3-none-any.whl (24 kB)
Using cached sqlparse-0.5.5-py3-none-any.whl (46 kB)
Installing collected packages: sqlparse, asgiref, django
Successfully installed asgiref-3.11.1 django-5.2.11 sqlparse-0.5.5
PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice>
```

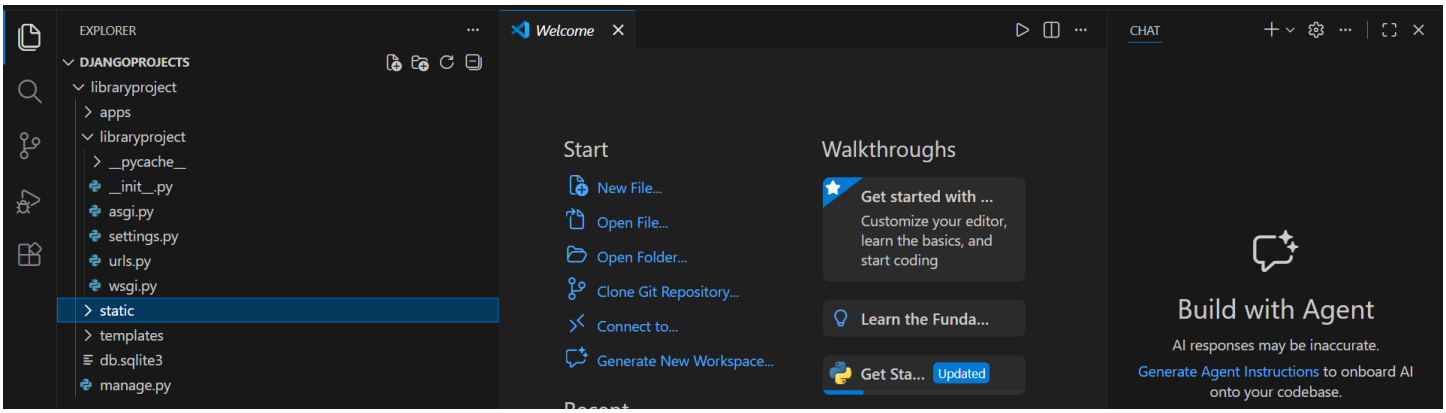
Task 2: create, configure, and organize a new Django project

```
PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice> mkdir DjangoProjects


    Directory: C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          3/2/2026   3:43 AM                DjangoProjects


PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice> cd DjangoProjects
PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice\DjangoProjects> django-admin startproject libraryproject
PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice\DjangoProjects> cd libraryproject
PS C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice\DjangoProjects\libraryproject> dir


    Directory: C:\Users\re2na\OneDrive\حطس \بتكمل\T_W\practice\DjangoProjects\libraryproject


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          3/2/2026   3:44 AM                libraryproject
-a----          3/2/2026   3:44 AM            692 manage.py
```

```
PS C:\Users\re2na\OneDrive\طس> بتكملا\T_W\practice\DjangoProjects\libraryproject> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, conten
ttypes, sessions.
Run 'python manage.py migrate' to apply them.
March 02, 2026 - 03:46:57
Django version 5.2.11, using settings 'libraryproject.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/
[02/Mar/2026 03:47:18] "GET / HTTP/1.1" 200 12068
```

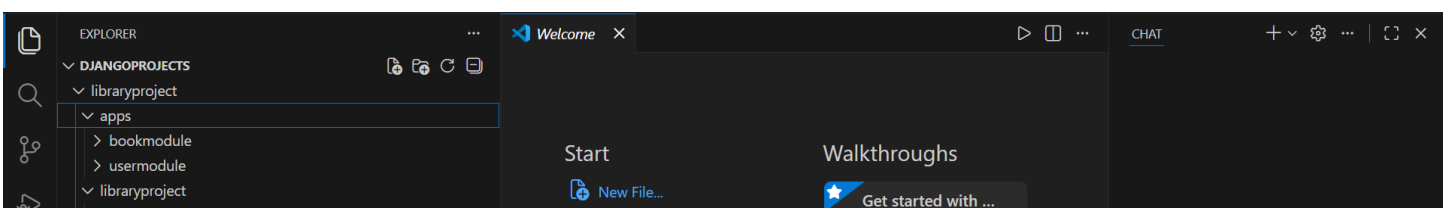


Task 3: configure your Django application

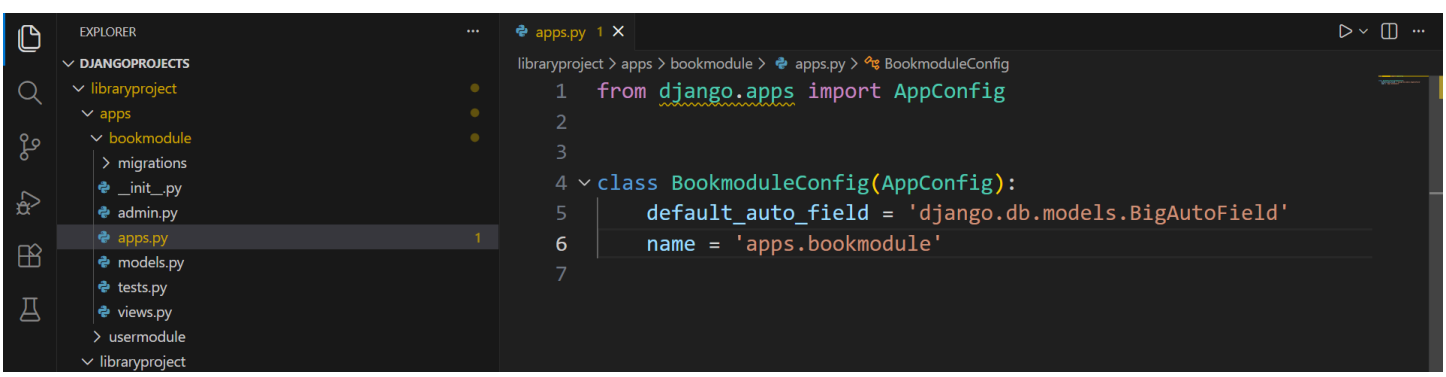


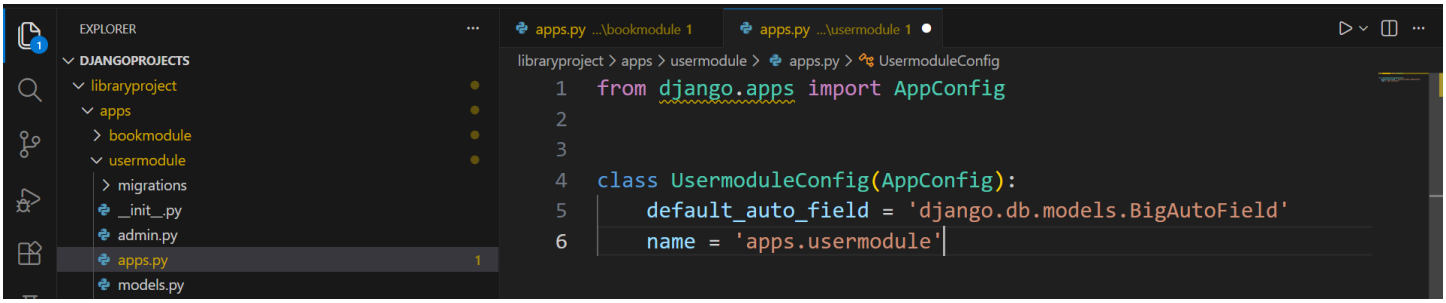
Step 2: create two module services in the apps folder: bookmodule and usermodule

```
(djangoenv) PS C:\Users\re2na\OneDrive\حطب \بتكمل\T_W\practice\DjangoProjects\libraryproject> cd apps
(djangoenv) PS C:\Users\re2na\OneDrive\حطب \بتكمل\T_W\practice\DjangoProjects\libraryproject\apps> python ../manage.py startapp bookm
odule
(djangoenv) PS C:\Users\re2na\OneDrive\حطب \بتكمل\T_W\practice\DjangoProjects\libraryproject\apps> python ../manage.py startapp userm
odule
(djangoenv) PS C:\Users\re2na\OneDrive\حطب \بتكمل\T_W\practice\DjangoProjects\libraryproject\apps>
```

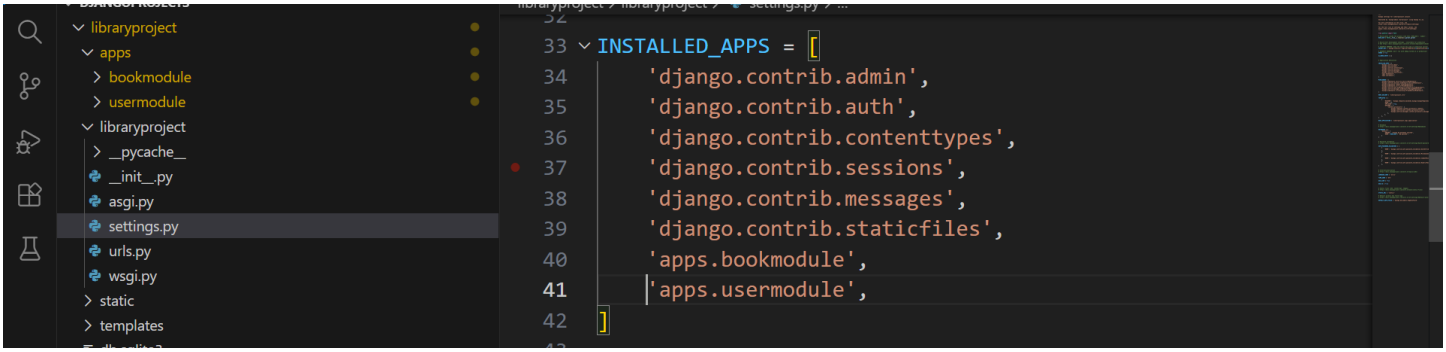


Step 3:





Step 4:



Step 5:

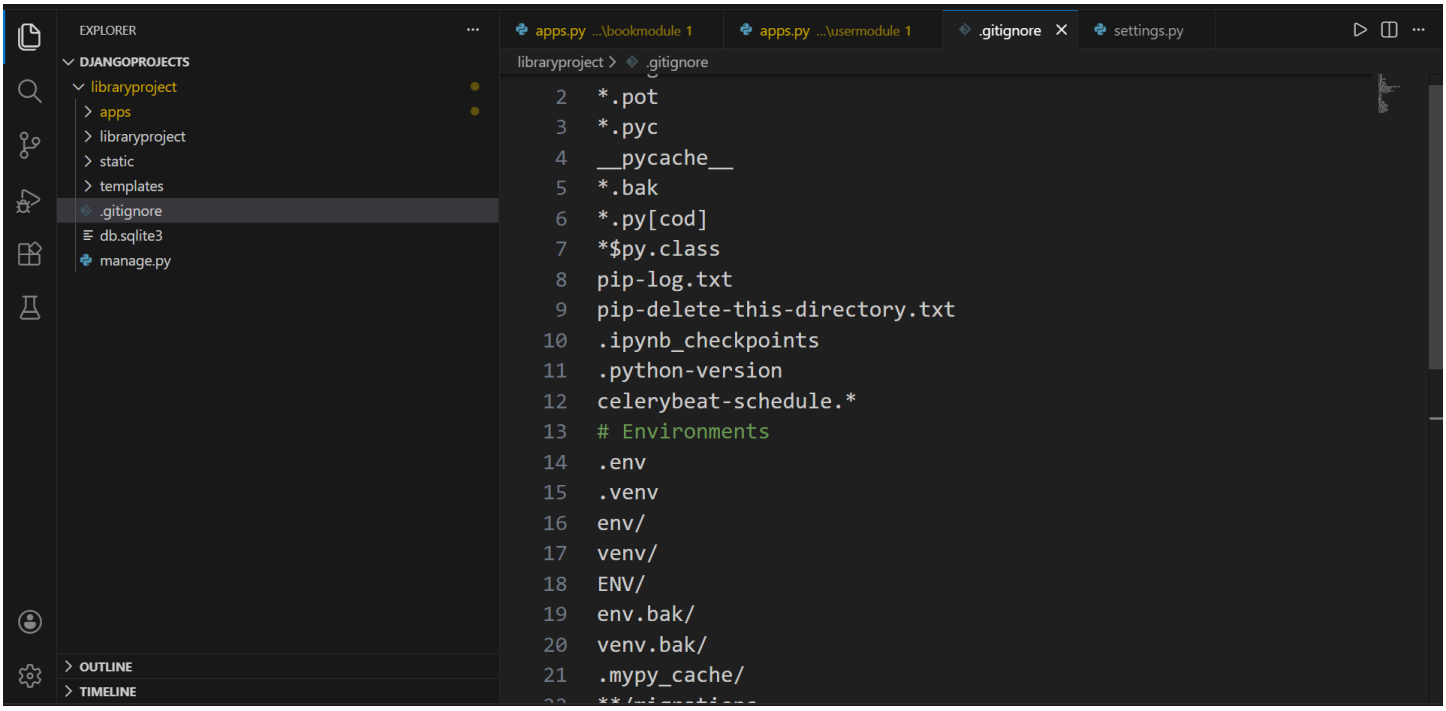
```
(djangoenv) PS C:\Users\re2na\OneDrive\حطس بتكمل\T_W\practice\DjangoProjects\libraryproject> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, conten
ttypes, sessions.
Run 'python manage.py migrate' to apply them.
March 02, 2026 - 04:26:34
Django version 5.2.11, using settings 'libraryproject.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/
[02/Mar/2026 04:26:51] "GET / HTTP/1.1" 200 12068
Not Found: /usermodule/
[02/Mar/2026 04:27:20] "GET /usermodule/ HTTP/1.1" 404 2216
```

Part 2: Git system and connect your online repo with a local repo

Task 1: Create a local git repo.

Step 1:

Step 3:




```
(djangoenv) PS C:\Users\re2na\OneDrive\حطس ٨بتكمل\T_W\practice\DjangoProjects\libraryproject> git init
Initialized empty Git repository in C:/Users/re2na/OneDrive/حطس ٨بتكمل/T_W/practice/DjangoProjects/libraryproject/.git/
(djangoenv) PS C:\Users\re2na\OneDrive\حطس ٨بتكمل\T_W\practice\DjangoProjects\libraryproject> git add .gitignore
(djangoenv) PS C:\Users\re2na\OneDrive\حطس ٨بتكمل\T_W\practice\DjangoProjects\libraryproject> git commit -m "add ignore file"
[master (root-commit) bd29e08] add ignore file
 1 file changed, 26 insertions(+)
 create mode 100644 .gitignore
(djangoenv) PS C:\Users\re2na\OneDrive\حطس ٨بتكمل\T_W\practice\DjangoProjects\libraryproject>
```

```
(djangoenv) PS C:\Users\re2na\OneDrive\حطس ٨بتكمل\T_W\practice\DjangoProjects\libraryproject> git add .gitignore
(djangoenv) PS C:\Users\re2na\OneDrive\حطس ٨بتكمل\T_W\practice\DjangoProjects\libraryproject> git commit -m "Lab 2 Django project set
up"
[master aee1a56] Lab 2 Django project setup
 19 files changed, 236 insertions(+), 1 deletion(-)
 create mode 100644 apps/bookmodule/__init__.py
 create mode 100644 apps/bookmodule/admin.py
 create mode 100644 apps/bookmodule/apps.py
 create mode 100644 apps/bookmodule/models.py
 create mode 100644 apps/bookmodule/tests.py
 create mode 100644 apps/bookmodule/views.py
 create mode 100644 apps/usermodule/__init__.py
 create mode 100644 apps/usermodule/admin.py
 create mode 100644 apps/usermodule/apps.py
 create mode 100644 apps/usermodule/models.py
 create mode 100644 apps/usermodule/tests.py
 create mode 100644 apps/usermodule/views.py
 create mode 100644 libraryproject/__init__.py
 create mode 100644 libraryproject/asgi.py
 create mode 100644 libraryproject/settings.py
 create mode 100644 libraryproject/urls.py
 create mode 100644 libraryproject/wsgi.py
 create mode 100644 manage.py
```

Task 2: connect your local repository with the remote repository you created in Lab 2.

```
(djangoenv) PS C:\Users\re2na\OneDrive\حطس\بتكمل\T_W\practice\DjangoProjects\libraryproject> git branch -M main
(djangoenv) PS C:\Users\re2na\OneDrive\حطس\بتكمل\T_W\practice\DjangoProjects\libraryproject> git remote add origin https://github.co
m/Reema2N/Web_Technologies_lab.git
(djangoenv) PS C:\Users\re2na\OneDrive\حطس\بتكمل\T_W\practice\DjangoProjects\libraryproject> git push -u origin main
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 12 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (22/22), 3.96 KiB | 676.00 KiB/s, done.
Total 22 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/Reema2N/Web_Technologies_lab.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
(djangoenv) PS C:\Users\re2na\OneDrive\حطس\بتكمل\T_W\practice\DjangoProjects\libraryproject>
```